CSE STUDY HUB

DATA COMMUNICATION & NETWORKING UNIT 1

Introduction to Computer Networks & Physical Layer

SOURCE-ALIGNED CONCISE NOTES

Prepared using the uploaded handwritten class notes for this unit. The explanations retain the concepts and examples visible in those notes, with wording cleaned up for revision.

Topics covered in this unit

- What the uploaded notes cover
- Network models and OSI
- Transmission media and physical-layer ideas
- Quick diagram cues and exam practice

How to use: Learn the definition, draw the diagram/flow where applicable, practise the numerical method, and finish with the revision questions. If a detail is not present in the uploaded pages, it is not added as an asserted class-note topic.

Exam answer structure

For a 5-mark answer: definition → labelled diagram/formula → working or components → example/application → advantage, limitation or comparison.

UNIT 1 · What the uploaded notes cover

Computer networks connect computers and smart devices so that they can exchange information and share resources. Examples in the notes include computers, smartphones, smart watches and other connected devices.

Communication depends on end devices, a communication medium and network equipment such as switches and routers. A protocol defines the rules followed by communicating devices.

Network size and coverage: PAN is a personal-area network, LAN covers a limited local area, MAN spans a metropolitan area, and WAN covers a large geographical area.

Common network arrangements include client–server and peer-to-peer. In client–server, a server provides services to clients; in peer-to-peer, connected devices can share resources directly.

UNIT 1 · Network models and OSI

The OSI reference model divides communication into seven layers. From bottom to top: Physical, Data Link, Network, Transport, Session, Presentation and Application.

Physical: transmits raw bits as electrical, optical or radio signals. Data Link: organizes bits into frames and uses MAC addressing for local delivery. Network: handles logical addressing and routing between networks.

Transport: provides process-to-process delivery using ports and transport protocols. Session: manages communication sessions. Presentation: handles data representation, translation, compression and encryption. Application: provides network services to user applications.

Layering makes a complex system easier to design, troubleshoot and standardize. Each layer uses services from the layer below and provides services to the layer above.

UNIT 1 · Transmission media and physical-layer ideas

Guided media carry signals through a physical path: twisted-pair cable, coaxial cable and optical fibre. Unguided media transmit through air or space using radio, microwave or infrared waves.

Bandwidth describes the capacity of a communication channel. Bit rate is the number of bits transmitted per second. Transmission delay depends on the amount of data and the link bit rate.

Manchester encoding represents each bit using a transition in the middle of the bit interval. The direction of the transition distinguishes 0 from 1; the transition also helps the receiver maintain timing.

Transmission can be simplex (one direction), half-duplex (both directions, but not simultaneously) or full-duplex (both directions simultaneously).

UNIT 1 · Quick diagram cues and exam practice

Draw the OSI stack as seven labelled horizontal layers. Add one example responsibility beside each layer; do not reverse the order.

Draw a simple network: two end devices connected through a switch, then a router connecting to another LAN. Label the local link and the inter-network path.

Compare twisted pair, coaxial cable and fibre by signal type, interference resistance, distance and typical use. Compare simplex, half-duplex and full-duplex with one example each.

Revision questions: (1) Explain the seven OSI layers. (2) Differentiate LAN, MAN and WAN. (3) Compare guided and unguided media. (4) Explain Manchester encoding. (5) Differentiate simplex, half-duplex and full-duplex.